Yuan Gong

Incoming Assistant Professor
Johns Hopkins University
email: yuangongfdu@gmail.com
web: yuangongnd.github.io

Bio

I am an audio and speech researcher and engineer with the long-term goal of building machines that perceive, understand, and reason about the world through sound. My work spans multi-modal large language models, audio-visual learning, speech AI for health, and trustworthy speech systems.

Education

Fudan University	Biomedical Engineering	B.S. (2015)
University of Notre Dame	Computer Science and Engineering	Ph.D. (2020)

Appointments

Aug 2024 – Apr 2026, Member of Technical Staff and Senior Manager, xAI, Palo Alto, CA
Aug 2023 – Aug 2024, Research Scientist II, MIT CSAIL, Cambridge, MA
Aug 2020 – Jul 2023, Postdoctoral Associate, MIT CSAIL, Cambridge, MA
May 2019 – Aug 2019, Applied Scientist Intern, Amazon Web Services, Seattle, WA
Aug 2015 – Jul 2020, Research Assistant, University of Notre Dame, Notre Dame, IN

Honors

- ASRU Best Paper Finalist, 12 of 435 papers (2023)
- ICASSP Outstanding Reviewer (2023)
- Interspeech Best Student Paper Award Nomination (2019)
- AVEC Depression Detection Challenge, First Place (2017)
- Fudan First Prize Scholarship and Outstanding Graduate (2015)

Five Representative Papers

- Y. Gong, H. Luo, A. H. Liu, L. Karlinsky, J. Glass. "Listen, Think, and Understand," ICLR, 2024.
- Y. Gong, A. Rouditchenko, A. H. Liu, D. Harwath, L. Karlinsky, H. Kuehne, J. Glass. "Contrastive Audio-Visual Masked Autoencoder," ICLR, 2023.
- Y. Gong, Y.-A. Chung, J. Glass. "AST: Audio Spectrogram Transformer," Interspeech, 2021.
- Y. Gong, J. Yang, J. Huber, M. MacKnight, C. Poellabauer. "ReMASC: Realistic Replay Attack Corpus for Voice Controlled Systems," Interspeech, 2019.
- Y. Gong, C. Poellabauer. "Topic Modeling Based Multi-Modal Depression Detection," AVEC, ACM Multimedia, 2017.

Professional Service

Area Chair: ICLR 2025; NeurIPS 2026
Co-organizer: IEEE SLT 2024 Challenge (Generative Speech Transcription Error Correction); Interspeech 2024 Special Session (Responsible Speech Foundation Models)
Reviewer, over 70 papers: TPAMI, IEEE/ACM TASLP, IEEE TDSC, IEEE THMS, IEEE SPL, Springer Machine Learning, NeurIPS, ICLR, ICCV, Interspeech, ICASSP, DCASE
Committee Member: ACM-BCB 2019 Travel Grant Committee; M.S. Thesis Committee (Marisa Cameron, University of Notre Dame, 2017)

Invited Talks and Guest Lectures

- "From Audio Perception to Understanding: A Path Towards Audio AGI," Amazon GenAI (2024)

- "General Audio Processing: Perception, Understanding, and Generation," MIT 6.8620 / HST.728 Guest Lecture (2024)
- "How Do We Evaluate Audio and Speech Large Language Models?" ASRU Workshop on Speech Foundation Models (2023)
- "Audio Classification and Understanding at MIT SLS," Amazon Alexa Tutorial (2023)
- "Audio Large Language Models: From Sound Perception to Understanding," SANE Workshop; MIT Embodied Intelligence Seminar (2023)
- "Contrastive Audio-Visual Masked Autoencoder," HKUST (Guangzhou); IBM Watson AI Lab (2023)
- "Large Language Models that Listen," Signify Research; Takeda (2023)
- "Audio Spectrogram Transformer: Architecture, Training, and Pre-training," Adobe Research; ByteDance; MERL; AI Time; MIT EI Seminar; ISCA SIGML Seminar (2021–2022)
- "General Audio Processing," MIT 6.345 / HST.728 Guest Lecture (2022)
- "Security of Speech Processing Systems," University of Notre Dame CSE 60641 Guest Lecture (2018, 2019)

Journal Articles

- Y. Gong, S. Khurana, A. Rouditchenko, J. Glass. "CMKD: CNN/Transformer-Based Cross-Model Knowledge Distillation for Audio Classification," IEEE Transactions on Pattern Analysis and Machine Intelligence (TPAMI), 2026.
- Y. Gong, A. H. Liu, A. Rouditchenko, J. Glass. "UAVM: Towards Unifying Audio and Visual Models," IEEE Signal Processing Letters, 2022.
- Y. Gong, Y.-A. Chung, J. Glass. "PSLA: Improving Audio Tagging with Pretraining, Sampling, Labeling, and Aggregation," IEEE/ACM Transactions on Audio, Speech, and Language Processing, 2021.
- Y. Gong, J. Yang, C. Poellabauer. "Detecting Replay Attacks Using Multi-Channel Audio: A Neural Network-Based Method," IEEE Signal Processing Letters, 2020.
- Y. Gong, J. Cao, Z. Luo, G. Zhou. "A Smart Low-Power-Consumption ECG Monitor Based on MSP430F5529 and CC2540," Chinese Journal of Medical Instrumentation, 2015.

Conference Papers

- Y. Gong, H. Luo, A. H. Liu, L. Karlinsky, J. Glass. "Listen, Think, and Understand," ICLR, 2024.
- A. Rouditchenko, Y. Gong, S. Thomas, L. Karlinsky, H. Kuehne, R. Feris, J. Glass. "Whisper-Flamingo: Integrating Visual Features into Whisper for Audio-Visual Speech Recognition and Translation," Interspeech, 2024.
- L. Wang, Y. Gong, N. Dawalatabad, M. Vilela, K. Placek, B. Tracey, Y. Gong, A. Premasiri, F. Vieira, J. Glass. "Automatic Prediction of Amyotrophic Lateral Sclerosis Progression Using Longitudinal Speech Transformer," Interspeech, 2024.
- T. Zhang, J. Ge, H. Luo, Y.-S. Chuang, M. Gao, Y. Gong, X. Wu, Y. Kim, H. Meng, J. Glass. "Natural Language Embedded Programs for Hybrid Language Symbolic Reasoning," Findings of NAACL, 2024.
- H. Luo, Y.-S. Chuang, Y. Gong, T. Zhang, Y. Kim, X. Wu, H. Meng, J. Glass. "SAIL: Search-Augmented Instruction Learning," Findings of EMNLP, 2023.
- Y. Gong, A. H. Liu, H. Luo, L. Karlinsky, J. Glass. "Joint Audio and Speech Understanding," IEEE ASRU, 2023. (Best Paper Finalist)
- Y. Gong, S. Khurana, L. Karlinsky, J. Glass. "Whisper-AT: Noise-Robust Automatic Speech Recognizers Are Also Strong General Audio Event Taggers," Interspeech, 2023.
- Y. Gong, A. Rouditchenko, A. H. Liu, D. Harwath, L. Karlinsky, H. Kuehne, J. Glass. "Contrastive Audio-Visual Masked Autoencoder," ICLR, 2023. (Notable-Top-25%)
- J. Yang, B. Xia, J. Bailey, Y. Gong, J. Templeton, C. Poellabauer. "Improving Computational Efficiency of Voice Anti-Spoofing Models," IEEE MASS, 2023.
- N. Dawalatabad, Y. Gong, S. Khurana, R. Au, J. Glass. "Detecting Dementia from Long Neuropsychological Interviews," Findings of EMNLP, 2022.
- Y. Gong, Z. Chen, I.-H. Chu, P. Chang, J. Glass. "Transformer-Based Multi-Aspect Multi-Granularity Non-Native English Speaker Pronunciation Assessment," IEEE ICASSP, 2022.
- Y. Gong, J. Yu, J. Glass. "VocalSound: A Dataset for Improving Human Vocal Sounds Recognition," IEEE

ICASSP, 2022.

- Y. Gong, C.-I. J. Lai, Y.-A. Chung, J. Glass. "SSAST: Self-Supervised Audio Spectrogram Transformer," AAAI, 2022.
- Y. Gong, Y.-A. Chung, J. Glass. "AST: Audio Spectrogram Transformer," Interspeech, 2021.
- Y. Gong, B. Li, C. Poellabauer, Y. Shi. "Real-Time Adversarial Attacks," IJCAI, 2019.
- Y. Gong, J. Yang, J. Huber, M. MacKnight, C. Poellabauer. "ReMASC: Realistic Replay Attack Corpus for Voice Controlled Systems," Interspeech, 2019. (Best Student Paper Nomination)
- N. Xia, Y. Gong, Y. Zhang, C. Poellabauer. "Non-Local Second-Order Attention Networks for Person Re-Identification," IEEE ICCV, 2019.
- Y. Gong, C. Poellabauer. "Impact of Aliasing on Deep CNN-Based End-to-End Acoustic Models," Interspeech, 2018.
- Y. Gong, H. Yatawatte, C. Poellabauer, S. Schneider, S. Latham. "Automatic Autism Spectrum Disorder Detection Using Everyday Vocalizations Captured by Smart Devices," ACM-BCB, 2018.
- Y. Gong, C. Poellabauer. "Protecting Voice Controlled Systems Using Sound Source Identification Based on Acoustic Cues," ICCCN, 2018.
- Y. Gong, C. Poellabauer. "Continuous Assessment of Children's Emotional States Using Acoustic Analysis," IEEE ICHI, 2017.

Workshop Papers

- Y. Gong, C. Poellabauer. "Crafting Adversarial Examples for Speech Paralinguistics Applications," DYNAMICS Workshop, ACSAC, 2018.
- Y. Gong, K. Shin, C. Poellabauer. "Improving LIWC Using Soft Word Matching," ACM-BCB (Poster), 2018.
- Y. Gong, C. Poellabauer. "An Overview of Vulnerabilities of Voice Controlled Systems," Workshop on Security and Privacy for IoT, 2018.
- Y. Gong, C. Poellabauer. "Topic Modeling Based Multi-Modal Depression Detection," AVEC, ACM Multimedia, 2017. (Challenge Winner)

Open-Source Software

Over 3,700 stars on GitHub: Audio Spectrogram Transformer (AST); Listen, Think, and Understand (LTU); SSAST; Whisper-AT; CAV-MAE; GOPT; VocalSound; PSLA; UAVM; ReMASC.

Media Coverage

"Scaling Audio-Visual Learning without Labels," MIT News (2023). Featured on AK / Hugging Face Daily Papers: AST (2021), SSAST (2021), VocalSound (2022), Whisper-AT (2023).

Students and Mentees

- Ph.D. students collaborated with at MIT: Yu-An Chung, Cheng-I Lai, Sameer Khurana, Alexander H. Liu, Andrew Rouditchenko
- M.S. students at Notre Dame: Marisa Cameron (2016–2017, committee member); Yu Jiang (2020–2021)
- Undergraduate students at Notre Dame: Michael Parowski, Jorge Diaz-Ortiz, John Considine, Kevin Shin, Royce Branning, Jacob Huber, Mitchell MacKnight, Jorge Jose Daboub Silhy, John Bailey